Suicide-Risk Detection Using NLP-Based Text and Speech Analysis

Complete Project Implementation Document

NLP + Speech Analytics + Multi-Signal Risk Screening

Presenting Team

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Guide

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Implementation architecture aligned with the approved Review-1 system design and the fully audited project datasets.

1. Project Objective

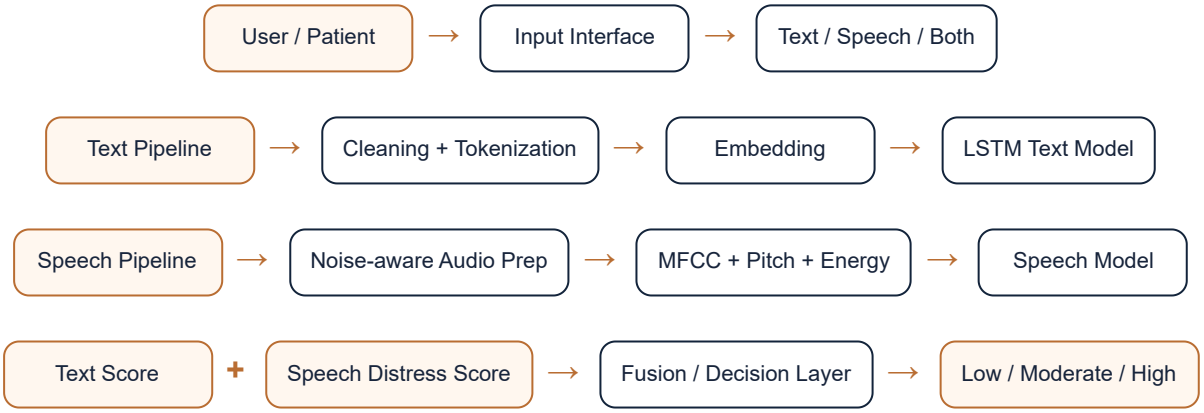
The project implements an NLP-centered mental-health screening prototype that analyzes **text**, **speech**, or both. The aim is to identify concerning language and vocal distress signals, then combine model outputs into a final screening-oriented risk result.

> **Core architecture rule:** text and speech are processed through separate pipelines first. Their outputs are combined only at the decision/fusion layer. This matches the Review-1 architecture.

Supported Input Modes

Mode	Input	Processing	Output
Text-only	User-entered text	NLP preprocessing → sequence model	Suicide-text risk probability
Speech-only	Recorded/uploaded voice	Audio preprocessing → acoustic features → speech model	Depression/distress probability
Text + Speech	Both modalities	Separate branch inference → calibrated fusion	Combined screening risk level

2. Final System Architecture



3. Datasets Used and Why

3.1 SDCNL / Reddit Text Dataset

Purpose: primary NLP branch for suicide-related text classification.

Why selected: the dataset already contains a direct binary target `is_suicide`, making it appropriate for supervised text classification of suicide-related language patterns.

Final File	Rows	Class 0	Class 1	Use
training-set-clean.csv	1513	727	786	Model training
testing-set-clean-leakage-free.csv	379	186	193	Final text-model testing
combined-set-clean.csv	1892	913	979	EDA/reference only

> **Important:** `combined-set-clean.csv` must not be trained together with the existing train/test split because it represents the combined source. Using it with the held-out test split would introduce leakage.

3.2 EATD-Corpus

Purpose: speech + transcript depression/distress signal branch.

Why selected: EATD provides paired audio, transcript text and SDS-based depression scores, allowing the speech branch to learn acoustic distress patterns such as MFCC, pitch, energy and related features.

Split	Participant Folders	Usable Participants	Depressed	Non-Depressed
Train	83	80	19	61
Validation	79	79	11	68

> **Scientific distinction:** EATD is a depression/distress dataset, not a direct suicide-risk-labelled dataset. Therefore, EATD is not merged row-by-row with SDCNL. It supplies a complementary speech distress score at the final decision layer.

4. Dataset Audit and Preprocessing Status

4.1 Text Dataset Processing

- Removed unwanted unnamed index columns.
- Selected `megatext_clean` as the source for final `model_text`.
- Normalized whitespace only; text meaning was not rewritten.
- Removed exact duplicate `model_text + is_suicide` records.
- Kept valid conflicting-label records without forced relabeling.
- Verified labels remain only 0 and 1.
- Completed train-test leakage audit: **0 overlapping normalized model_text records**.

4.2 EATD Processing

Expected files per valid participant:

- `positive.wav`, `positive_out.wav`, `positive.txt`
- `neutral.wav`, `neutral_out.wav`, `neutral.txt`
- `negative.wav`, `negative_out.wav`, `negative.txt`
- `label.txt` = raw SDS score
- `new_label.txt` = standardized SDS score

Implementation label rule for the speech branch:

> **depression_label = 1 if standardized SDS score ≥ 53, otherwise 0**

Known EATD Anomalies Handled

Location	Issue	Handling
Train participants 1, 10, 101	Empty folders	Excluded from feature rows; no labels invented
Train participant 102	label.txt missing	new_label.txt retained; raw score left missing
Validation participant 79 positive_out.wav	Zero-duration processed audio	Fallback to valid raw positive.wav

5. Text NLP Pipeline

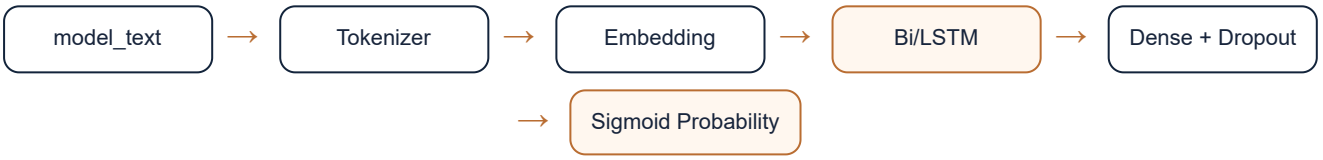
5.1 Input

Final text model input column: `model_text`
Final target: `is_suicide`

5.2 Preprocessing

- Whitespace normalization.
- Tokenizer fitting on training data only.
- Convert text into integer token sequences.
- Pad/truncate sequences to a fixed maximum length.
- Do not fit tokenizer on test data.

5.3 Planned Text Model



Primary implementation target is an LSTM-based sequence classifier, consistent with the proposed Review-1 architecture. A transformer model may later be used as a comparison baseline, but it does not replace the primary LSTM implementation.

5.4 Text Output

The branch produces a probability `P_text` from 0 to 1 representing the learned likelihood of suicide-related text patterns.

6. Speech Processing Pipeline

6.1 Audio Selection Rule

- Prefer `*_out.wav` because it is the dataset-provided preprocessed audio.
- If a processed audio file is unusable or zero-duration, use the corresponding raw `*.wav` only as a documented fallback.
- For model feature extraction, load audio as mono at 16 kHz.
- Apply peak amplitude normalization.

6.2 Extracted Acoustic Features

Feature	Purpose
13 MFCC means + 13 MFCC standard deviations	Compact spectral/voice representation
Pitch mean + standard deviation	Fundamental frequency variation
RMS energy mean + standard deviation	Voice intensity/energy
Zero-crossing-rate mean + standard deviation	Signal noisiness and temporal changes
Spectral centroid mean + standard deviation	Spectral brightness distribution
Duration	Utterance-length cue

6.3 Generated Feature Files

- `Processed_EATD\EATD-train-features.csv` — 240 audio-level rows
- `Processed_EATD\EATD-validation-features.csv` — 237 audio-level rows
- `Processed_EATD\EATD-train-participant-features.csv` — 80 participant rows
- `Processed_EATD\EATD-validation-participant-features.csv` — 79 participant rows

> **Recommended for modeling:** participant-level feature files. This keeps all three emotional recordings of one person together and avoids treating them as independent people.

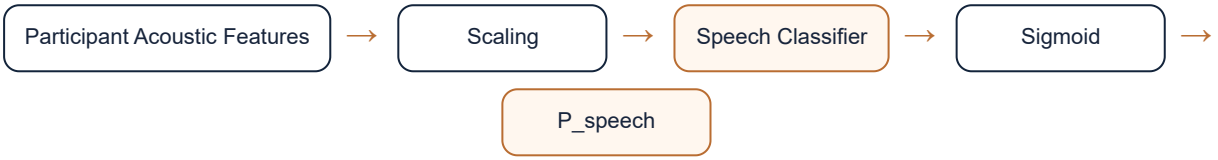
7. Speech Model

7.1 Input

Participant-level acoustic feature vectors containing negative, neutral and positive speech features.

7.2 Primary Model Strategy

Because the current speech representation is a fixed engineered feature vector, the first reliable implementation will use a compact dense neural network or another supervised classifier on participant-level features. An LSTM/CNN sequence model can be introduced when frame-level temporal speech sequences are retained.



7.3 Output

The branch produces `P_speech`, a depression/distress probability. This is a supporting risk signal, not a suicide diagnosis.

8. Fusion / Decision Layer

The project architecture requires text-only, speech-only and combined operation. Because SDCNL and EATD do not share the same participant identities or the same ground-truth target, training a single early-fusion model directly across both datasets would be scientifically invalid.

The correct implementation is **late decision fusion**:

Combined Score = w_1 × P_text + w_2 × P_speech
Initial prototype weights are tuned on development experiments rather than assumed permanently.

Risk categories are derived only at the final application layer:

- **Low:** lower combined score
- **Moderate:** intermediate combined score
- **High:** higher combined score

> Thresholds will be calibrated from validation behavior and documented. They will not be presented as clinical diagnostic cutoffs.

9. Model Training Procedure

Text Branch

1. Load `training-set-clean.csv`.
2. Fit tokenizer on training text only.
3. Create padded sequences.
4. Train LSTM model with validation split taken only from training data.
5. Save tokenizer and trained model.
6. Evaluate once on `testing-set-clean-leakage-free.csv`.

Speech Branch

1. Load `EATD-train-participant-features.csv`.
2. Separate metadata, label and acoustic feature columns.
3. Fit scaler on training features only.
4. Train speech classifier.
5. Evaluate on `EATD-validation-participant-features.csv`.
6. Save scaler and trained speech model.

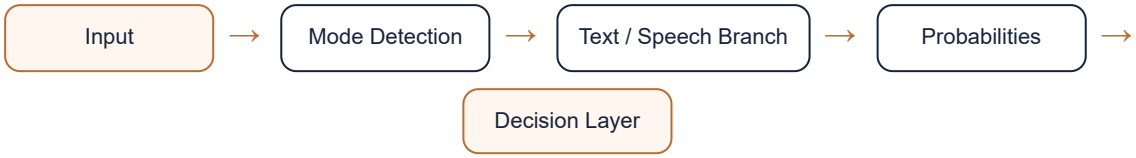
Reproducibility Controls

- Set Python/NumPy/framework random seeds.
- Save final hyperparameters.
- Do not tune using held-out text test data.
- Preserve participant separation in the EATD split.
- Version model artifacts and preprocessing scripts.

10. Evaluation Metrics

Metric	Reason
Accuracy	Overall correct prediction proportion
Precision	How many predicted positive cases are actually positive
Recall	How many actual positive cases are identified
F1-score	Balance between precision and recall
Confusion Matrix	Shows TP, TN, FP and FN behavior
ROC-AUC (recommended)	Threshold-independent probability ranking quality

11. Application Workflow



Text-only: display text-model screening result.
Speech-only: display distress-oriented speech result.
Both: show combined late-fusion risk level and supporting branch scores.

Expected Output Screen

- Input type detected.
- Text risk score, when text is available.
- Speech distress score, when speech is available.
- Final screening level: Low / Moderate / High.
- Short non-diagnostic support guidance.
- Clear notice that the result is a research screening prototype.

12. Proposed Project Folder Structure

```
Thanmayi 8000/
├── Datasets/
│   ├── training-set-clean.csv
│   ├── testing-set-clean-leakage-free.csv
│   ├── combined-set-clean.csv
│   ├── train/
│   ├── validation/
│   ├── Processed_EATD/
│   └── FINAL_DATASET_AUDIT_AND_USAGE.txt
├── src/
│   ├── text_preprocessing.py
│   ├── train_text_model.py
│   ├── speech_preprocessing.py
│   ├── train_speech_model.py
│   ├── fusion.py
│   └── inference.py
├── models/
│   ├── text_model/
│   └── speech_model/
├── app/
│   └── user_interface
├── reports/
│   ├── metrics/
│   └── confusion_matrices/
└── documentation/
```

13. Technology Stack

Area	Technology
Programming	Python 3.x
Data	Pandas, NumPy
NLP	Tokenizer, embeddings, TensorFlow/Keras or PyTorch LSTM
Speech	librosa, SciPy, wave/audio utilities
ML Evaluation	scikit-learn
Visualization	Matplotlib
Application	Flask / Streamlit / similar Python interface
Artifacts	Saved tokenizer, scaler, model weights, reports

14. Implementation Phases

Phase	Work	Status
1	Dataset selection and audit	COMPLETE
2	Text preprocessing and leakage removal	COMPLETE
3	EATD speech audit and feature extraction	COMPLETE
4	Text LSTM training and evaluation	NEXT
5	Speech model training and evaluation	PENDING
6	Fusion / decision layer	PENDING
7	User interface and inference pipeline	PENDING
8	Final testing, metrics, report and demo	PENDING

15. Critical Implementation Rules

- Do not mix `combined-set-clean.csv` into the final train/test experiment.
- Do not call EATD a suicide-labelled dataset.
- Do not directly concatenate SDCNL and EATD rows for supervised training.
- Do not invent labels for missing EATD files.
- Do not fit tokenizer/scaler on test or validation data.
- Keep model output framed as screening support, not clinical diagnosis.
- Save every preprocessing and model artifact needed to reproduce the result.

> **Current project status:** the complete dataset phase is ready. The next implementation action is to build and train the text LSTM model using the cleaned, leakage-free SDCNL split.

16. Final Implementation Direction

The project will remain aligned to the approved Review-1 architecture: **NLP text processing + speech feature processing + sequential/deep-learning models + multimodal decision fusion + risk classification + evaluation**.

For scientific correctness, the implementation separates direct suicide-text evidence from depression/distress speech evidence and combines them only as distinct screening signals. This preserves the intended multimodal architecture while avoiding false claims about dataset labels.

END OF IMPLEMENTATION DOCUMENT